

H14 round 1 (exploratory, non-holdout, 2026-10-03): entropy production of behavior-state sequences

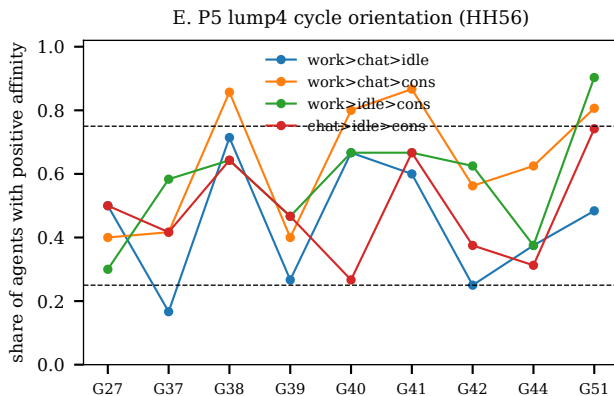
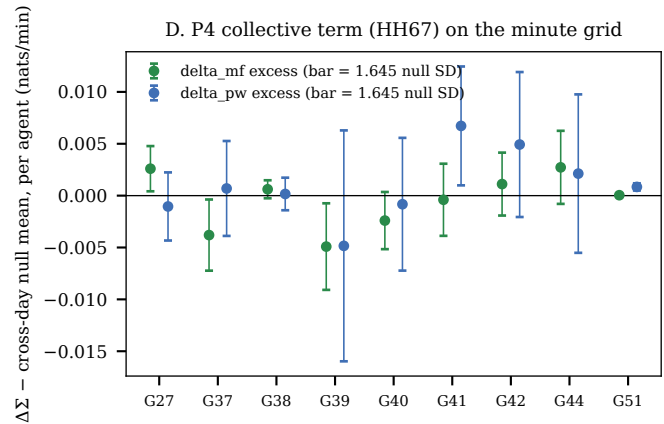
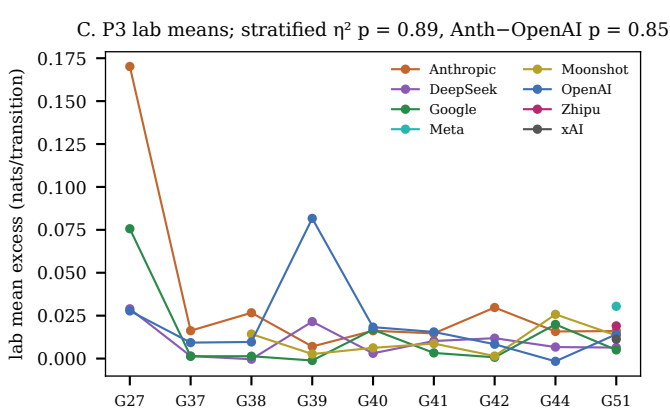
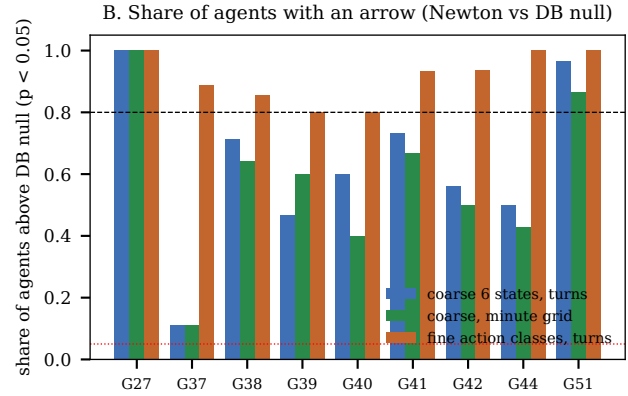
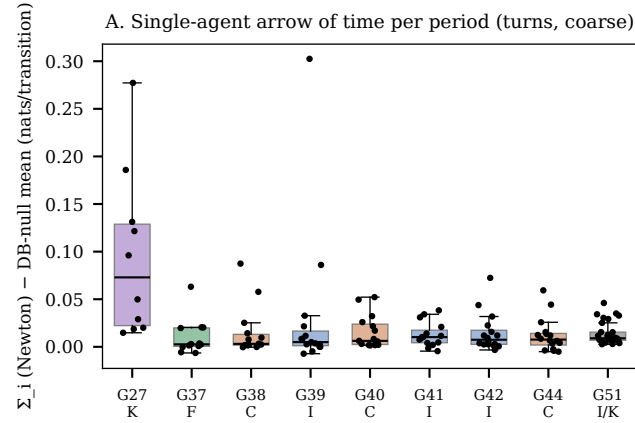
P2 single-agent arrow: holds ($\geq 80\%$ of agents above the detailed-balance surrogate) in 1/8 regime-III periods; per-period medians of Σ_i (Newton) 0.002-0.010 nats/transition.

P3 per-family (HH19): stratified η^2 $p = 0.886$; Anthropic – OpenAI summed over 8 periods = -0.0130 (two-sided $p = 0.854$, one-sided Anth>OpenAI $p = 0.585$; $I^2 = 0.13$). Agent trait stability: mean pairwise Spearman 0.13 (null 0.00, $p = 0.021$, 15 agents).

P4 collective (HH67): $\Delta\Sigma_{MF}$ above the cross-day null in no regime-III period (Holm-adjusted p : G37 1.00, G38 1.00, G39 1.00, G40 1.00, G41 1.00, G42 1.00, G44 0.87, G51 1.00).

P5 currents (HH56): holds in 0/8 regime-III periods. P6 regime contrast: True. P7: G37 lowest = True, mode C > mode I = False.

Verdicts: G27 mixed, G37 failed, G38 failed, G39 failed, G40 failed, G41 failed, G42 failed, G44 failed, G51 mixed



F. Synthetic validation (axis F)

S1 turn-like chains, $n=2,000 / 5$ d: $\Sigma=0$ bias Newton +0.0003, plug-in +0.0101; recovery at $\Sigma=0.06$: Newton 1.08, cfx-null 1.13; at $\Sigma=0.9$: Newton 1.06, cfx-null 0.99.

S1 minute-like chains, $n=2,000 / 5$ d: $\Sigma=0$ bias Newton -0.0004, plug-in +0.0098; recovery at $\Sigma=0.06$: Newton 0.94, cfx-null 0.99; at $\Sigma=0.7$: Newton 0.73, cfx-null 0.95.

S1n DB-surrogate test size at $\Sigma=0$ (8 cells, cold and stationary starts): Newton 0.027-0.067; power at $\Sigma=0.02$ ($n=2,000$): 0.95-1.0.

S2c family test (12 agents 6/4/2; equal true EP; n and stickiness differ): size Newton excess 0.04, raw plug-in 0.20, raw cfx 0.37; power at $2\times/3\times$: 0.32/0.66 (true values 0.40/0.71; agent heterogeneity limits power).

S3 kinetic Potts $N=15$, 5 d $\times$ 240 (cross-day null, reject rate MF / PW): independent 0.03/0.07 (true -0.00); shared field 0.00/0.03; coupled 1.00/1.00 (true 1.73); coupled_weak 0.93/1.00 (true 0.51); coupled_0.1 0.05/0.25 (true 0.02); coupled_0.2 0.20/0.25 (true 0.09).